

1. What is Python?

Python is:

- A) A compiled language
- B) A high-level, interpreted language
- C) A low-level language
- D) A markup language

2. Which of the following is a feature of Python?

- A) It is case-insensitive
- B) It supports object-oriented programming
- C) It does not support dynamic typing
- D) It cannot be used for AI/ML

3. Python is best suited for which of the following applications?

- A) Web Development
- B) Data Science
- C) Machine Learning
- D) All of the above

4. What is the default extension of a Python file?

- A) .py
- B) .python
- C) .p
- D) .txt

5. Which statement is used to output data in Python?

- A) console.log()
- B) cout
- C) print()
- D) echo

6. Which of the following is the correct way to declare a variable in Python?

- A) var x = 10

B) `int x = 10`

C) `x = 10`

D) `x := 10`

7. What is the correct way to take input from the user?

A) `input()`

B) `scan()`

C) `get()`

D) `cin`

8. Which of the following is true about Python indentation?

A) It is optional

B) It is used for comments

C) It is used to define blocks of code

D) It has no significance

9. What will be the output of `print(2 ** 3)`?

A) 6

B) 8

C) 9

D) 16

10. How are comments written in Python?

A) `// This is a comment`

B) `# This is a comment`

C) `/* This is a comment */`

D) `-- This is a comment`

11. Which of the following is not a valid variable name?

A) `my_var`

B) `1var`

C) `myVar`

D) `_var`

12. How do you define a multi-line comment in Python?

- A) /* comment */
- B) // comment
- C) ''' comment '''
- D) # comment #

13. Which operator is used for floor division?

- A) /
- B) %
- C) //
- D) **

14. What will be the output of `print(type(5.0))`?

- A) <class 'int'>
- B) <class 'float'>
- C) <class 'double'>
- D) <class 'number'>

15. How do you perform explicit type conversion in Python?

- A) `str(10)`
- B) `convert(str, 10)`
- C) `toString(10)`
- D) `string(10)`

16. What is the difference between Python 2 and Python 3?

- A) Python 2 uses `print` as a function, while Python 3 uses `print` as a statement
- B) Python 2 does not support integer division
- C) Python 3 supports Unicode by default
- D) Python 3 does not support exception handling

17. Which of the following is an identifier in Python?

- A) `my_var`
- B) `123var`
- C) `for`
- D) `import`

18. What is the use of the is operator?

- A) To compare values
- B) To check identity
- C) To perform bitwise operations
- D) To assign values

19. Which of the following is a Python keyword?

- A) class
- B) myClass C) define
- D) variable

20. Which function is used to determine the data type of a variable?

- A) typeof()
- B) type()
- C) dataType()
- D) checkType()

21. What will be the output of print(bool(0))?

- A) True
- B) False
- C) 0
- D) 1

22. What is an implicit type conversion in Python?

- A) Manually converting data types
- B) Python automatically converts types
- C) Using int() function
- D) Using float() function

23. Which symbol is used for single-line comments in Python?

- A) //
- B) #
- C) /* */
- D) --

24. Which function is used to get user input in Python 3?

- A) input()
- B) scan()
- C) get()
- D) cin

25. Which of the following is a valid literal in Python?

- A) Integer
- B) Float
- C) String
- D) All of the above

26. What is the output of `print(5 / 2)`?

- A) 2
- B) 2.5
- C) 2.0
- D) 3

27. What is the output of `print(5 // 2)`?

- A) 2.5
- B) 2
- C) 2.0
- D) 3

28. Which of the following is not a Python data type?

- A) int
- B) float
- C) real
- D) str

30. Which of the following statements is correct about Python?

- A) Python is a compiled language
- B) Python is an interpreted language
- C) Python is both compiled and interpreted
- D) Python is neither compiled nor interpreted

34. What is the output of `bool([])`?

- A) True
- B) False
- C) None
- D) Error

35. What is the output of `print(3 * "Hello")`?

- A) HelloHelloHello
- B) Hello * 3
- C) Error
- D) Hello Hello Hello

36. What will be the output of `print(10 == 10.0)`?

- A) True
- B) False
- C) Error
- D) None

37. What is the result of `10 % 3`?

- A) 3
- B) 1
- C) 0
- D) 10

38. What is the output of `print(type("Hello"))`?

- A) `<class 'str'>`
- B) `<class 'char'>`
- C) `<class 'text'>`
- D) `<class 'word'>`

39. What will be the output?

`a = 10`

`b = 3`

`print(a // b)`

What does this return?

```
print(5 + 2 * 3)
```

40. What will `print(x)` output?

```
x = 5
```

```
x += 3
```

```
print(x)
```

41. What type of operator is `**` in Python?

```
print(2 ** 3)
```

What will `print(y)` output?

```
y = 10
```

```
y %= 4
```

```
print(y)
```

42. What is the output of this comparison?

```
print(10 == 10.0)
```

43. What does the following print?

```
print(10 != 5 and 2 < 3)
```

Predict the output:

```
x = True
```

```
y = False
```

```
print(x or y and x)
```

44. What is the result of this expression?

```
print(5 > 3 > 1)
```

45. What will `print(a is b)` output?

```
a = [1, 2, 3]
```

```
b = [1, 2, 3]
```

```
print(a is b)
```

46. What does this bitwise operation return?

```
print(5 & 3)
```

Find the output:

```
print(8 | 2)
```

47. What does ~5 return?

```
print(~5)
```

Calculate using bitwise XOR:

```
print(7 ^ 2)
```

48. What is the result of shifting left?

```
print(3 << 2)
```

Variable Assignment & Scope

49. What will be the value of y after this code executes?

```
x = 10
```

```
y = x
```

```
x = 20
```

```
print(y)
```

50. What happens when you run this?

```
x = 5
```

```
def func():
```

```
    global x
```

```
    x = 10
```

```
func()
```

```
print(x)
```

51. What is the value of b?

```
a, b, c = 1, 2, 3
```

```
a, b = b, a
```

```
print(b)
```

52. What will print(x) output?

```
x = "Hello"
```

```
x += " World"
```

```
print(x)
```

Predict the output:

```
a = 5
```



```
b = a
```

```
a = a + 2
```

```
print(b)
```